Course: Full Stack Web Development
Course Code: 23CM4121
Task No.: 3
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Title

Develop To-Do list application

Aim

To develop a simple **To-Do List application** using HTML, CSS, JavaScript, DOM manipulation, and event listeners. The application allows users to add tasks, mark tasks as completed, delete tasks, and displays an appropriate message when no tasks are available.

OBJECTIVE

The objectives of this task are:

- To understand the basics of HTML, CSS, and JavaScript.
- To create the structure of a To-Do List application using HTML.
- To style the webpage using CSS properties.
- To manipulate HTML elements dynamically using JavaScript.
- To add new tasks dynamically to the webpage.
- To implement event listeners for button click events.
- To mark tasks as completed using CSS classes.
- To delete tasks dynamically from the DOM.
- To display an appropriate message when no tasks are available.
- To understand interactive webpage development using JavaScript.

THEORY

The **To-Do List application** is an interactive web application developed using **HTML, CSS, and JavaScript**.

HTML is used to create the basic structure of the webpage, including a text input field for entering tasks, an **Add Task** button, a task list area, and a message displaying **"No tasks available"** when the list is empty.

CSS is used to style the application by applying background colors, spacing, borders, rounded corners, shadows, and layouts. A CSS class named .completed is used to visually indicate that a task has been completed.

JavaScript is used for **DOM manipulation and event handling**. When the user enters a task and clicks the **Add Task** button, a new task element is dynamically created using createElement().

Each task contains a **Complete** button and a **Delete** button. The Complete button uses classList.toggle() to add or remove the completed class. The Delete button removes the corresponding task from the webpage using the remove() method.

When there are no tasks in the list, the **"No tasks available"** message is displayed again.

The basic flow of the application is:

User Input → Add Task → Create Task Element → Add to DOM → Complete/Delete Task → Update Task List

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ALGORITHM / PROCEDURE

1. Create an HTML file for the To-Do List application.
2. Add a heading titled **My To-Do List**.
3. Create a text input field for entering tasks.
4. Create an **Add Task** button.
5. Create a task list container.
6. Add a **"No tasks available"** message inside the task list.
7. Create a CSS file to style the webpage.
8. Style the input field, button, task list, and task items.
9. Define a .completed CSS class to show completed tasks using a line-through effect.
10. Create a JavaScript file.
11. Select the required HTML elements using getElementById().
12. Add a click event listener to the **Add Task** button.
13. Retrieve and trim the task entered by the user.
14. If the input is empty, display an alert message.
15. Otherwise, dynamically create a task item using createElement().

16. Create **Complete** and **Delete** buttons for the task.
17. Add a click event listener to the Complete button.
18. Toggle the completed class when the Complete button is clicked.
19. Add a click event listener to the Delete button.
20. Remove the task when the Delete button is clicked.
21. Display the **"No tasks available"** message when all tasks are deleted.
22. Clear the input field after adding a task.
23. Open the webpage in a browser.
24. Perform multiple test cases and verify the actual output.

PROGRAM

HTML Code:

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Document</title>
    <link rel="stylesheet" href="./style.css">
</head>
<body>
    <h1>My To-Do List</h1>
    <div class="container">
        <label for="task">Enter the Task:</label>
        <input type="text" name="" id="task" placeholder="Please enter your task">

        <br><br>

        <button id="addBtn">Add Task</button>

    </div><br><br>
    <div id="taskList">
        <p id="emptyMessage">No tasks available</p>
    </div>
    <script src="./script.js"></script>
</body>
</html>
```

CSS Code:

```css
body{
    background-color: blueviolet;
    display: flex;
    flex-direction: column;
    justify-content: center;
    align-items: center;
    gap: 10px;
}
.container{
    background-color: white;
    padding: 20px;
    border-radius: 8px;
    font-size: 1.3rem;
    box-shadow: 10px 20px 30px 10px rgba(0, 0, 0, 0.15);
}
#task{
    border-radius: 4px;
    border: none;
    background-color: rgb(183, 133, 231);
    height: 20px;
    padding: 5px;
}
#task::placeholder{
    color: white;
}
#addBtn{
    margin-left: 100px;
    cursor: pointer;
    padding: 10px;
    background-color: burlywood;
    border: none;
    border-radius: 4px;
}
#taskList{
    display: flex;
    flex-direction: column;
    gap: 10px;
    background-color: burlywood;
    padding: 10px;
    border-radius: 10px;
    box-shadow: 10px 20px 30px 10px rgba(0, 0, 0, 0.15);

}
.task-item {
    background-color: white;
    padding: 12px 15px;
    border-radius: 5px;

    display: flex;
    align-items: center;
    justify-content: space-between;
    gap: 10px;

    min-width: 300px;
}
.task-item button {
    padding: 6px 10px;
    cursor: pointer;
    border: none;
    border-radius: 4px;
    background-color: burlywood;
}
.completed{
    text-decoration: line-through;
    opacity: 0.6;
}
```

Javascript Code:

```javascript
const task = document.getElementById("task");
const addBtn = document.getElementById("addBtn");
const taskList = document.getElementById("taskList");
const emptyMessage = document.getElementById("emptyMessage");

addBtn.addEventListener("click", function(){
    const taskValue = task.value.trim()

    if(taskValue === ""){
        alert("Please Enter the task")
    }
    else{
        const taskItem = document.createElement("div")
        taskItem.textContent = taskValue;
        taskItem.classList.add("task-item");

        const completeBtn = document.createElement("button");
        completeBtn.textContent = "Complete";

        const deleteBtn = document.createElement("button");
        deleteBtn.textContent = "Delete";

       completeBtn.addEventListener("click", function(){
        taskItem.classList.toggle("completed");
       });

       deleteBtn.addEventListener("click", function(){
        taskItem.remove();

        if(taskList.children.length === 1){
            emptyMessage.style.display = "block";
        }
       });

       taskItem.appendChild(completeBtn);
       taskItem.appendChild(deleteBtn)

       taskList.appendChild(taskItem);

       emptyMessage.style.display = "none";

       task.value = "";
    }
});
```

Code Explanation

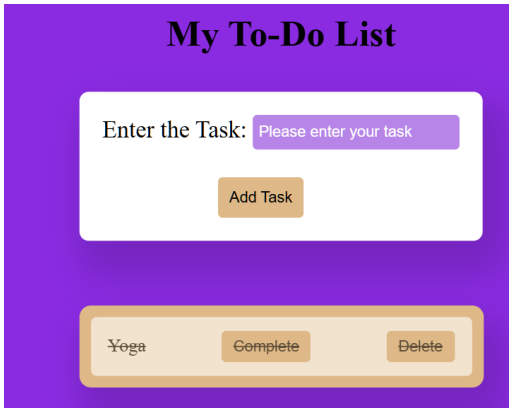
e	lanation
lementById()	cts an HTML element using its ID.
EventListener()	dles events such as button clicks.
ue	ieves the value entered in the input field.
()	oves unnecessary spaces from the input.
teElement()	ates a new HTML element dynamically.

Content	or retrieves the text content of an element.
sList.add()	s a CSS class to an element.
sList.toggle()	s or removes a CSS class dynamically.
endChild()	s a newly created element to another element.
ove()	oves an element from the webpage.
e.display	trols whether an element is visible or hidden.
pleted	lies line-through and reduced opacity to completed tasks.
askValue === "")	cks whether the task input is empty.
()	lays a message when the user tries to add an empty task.

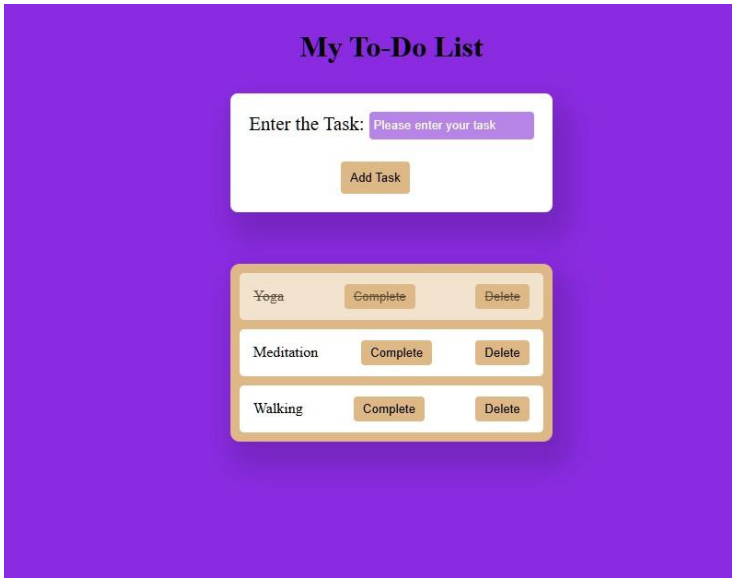
TEST CASES AND EXECUTION DATA

Case	ponent ed	Action / Input	Exected Result	Final Result	Status
01	Task	r "Complete gnment" and click Task	should be added e task list	added essfully	PASS
02	ty Task dation	ve the input field ty and click Add	t message "Please r the task" should isplayed	t displayed essfully	PASS
03	plete Task	a task and click plete	should be layed with line-ugh effect	marked as pleted essfully	PASS
04	ete Task	a task and click ete	cted task should emoved from the	deleted essfully	PASS
05	ty Task	ete all tasks	tasks available" sage should be layed	sage displayed essfully	PASS

TC OUTPUT 01 & 02:



OUTPUT:



RESULT

Thus, a To-Do List application was successfully developed using HTML, CSS, and JavaScript. DOM manipulation and event listeners were used to add, complete, and delete tasks dynamically. Empty task validation and the "No tasks available" message were also implemented successfully. The application was tested using multiple test cases and the expected results were verified.